Business Intelligence SQL Assignment Instructions

First of all, have a look at the SQL database creation and population code that accompanies this

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assignment. You should be familiar with the fields and what kinds of data are represented in the database. **All questions and items in this assignment pertain to the SQL code that accompanies this assignment.**

You must include all SQL code here in the spaces below, respecting standard capitalization conventions as presented in the tutorials. **For ease of review, please use BOLD on any SQL code (see the example below). Your SQL code must be in text format, so that the instructor can cut and paste for review and testing purposes. Here is an example of SQL code/text that can be cut and pasted for testing:**

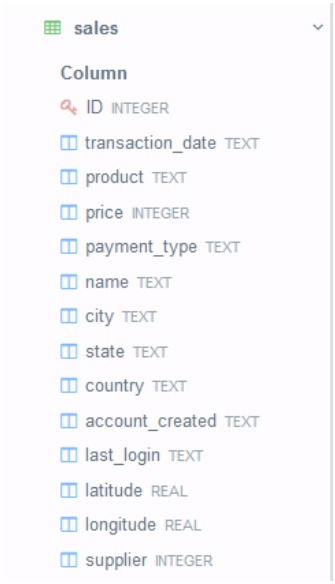
```
CREATE TABLE customers (id INTEGER PRIMARY KEY, name TEXT, age INTEGER, weight REAL);

INSERT INTO customers VALUES (73, "Brian", 33, 50);
```

For certain responses you are <u>also</u> asked to include a 'snip' / screenshot of your results. If you can, try not to take an image of the entire screen, but only of the relevant output.

You should use sqliteonline.com to complete this assignment, both to test code and take screenshots:

https://sqliteonline.com/

Here is an example of a database schema image and a query results image from sqliteonline.com. Due to screen resolution, it's not always possible to capture all query results. But be sure your images are legible, no smaller than the second image below. If an image is not legible, your assignment may be returned to you and you'll have to rerun the SQL scripts.



```
SELECT * FROM sales;
```

ID	transacti...	product	price	paymen...	name	city	state	country	account...	last_login	latitude	longitude	supplier
1	1/2/19 4:53	Chair	1200	Visa	Bettina S...	Parkville	MO	United St...	1/2/19 4:42	1/2/19 7:49	39.195	-94.68194	2
2	1/2/19 13:08	Chair	1200	Mastercard	Federica ...	Astoria	OR	United St...	1/1/19 16:21	1/3/19 12:32	46.18806	-123.83	3
3	1/4/19 12:56	Couch	3600	Visa	Gerd Win...	Cahaba H...	AL	United St...	11/15/18 ...	1/4/19 12:45	33.52056	-86.8025	1
4	1/4/19 13:19	Chair	1200	Visa	Laurence ...	Mickleton	NJ	United St...	9/24/18 1...	1/4/19 13:04	39.79	-75.23806	3
5	1/4/19 20:11	Chair	1200	Mastercard	Fleur Bar...	Peoria	IL	United St...	1/3/19 9:38	1/4/19 19:45	40.69361	-89.58889	2
6	1/2/19 20:09	Chair	1200	Mastercard	Adam Ma...	Martin	TN	United St...	1/2/19 17:43	1/4/19 20:01	36.34333	-88.85028	4
7	1/5/19 2:42	Chair	1200	Diners	Stacy Ro...	New York	NY	United St...	1/5/19 2:23	1/5/19 4:59	40.71417	-74.00639	2
8	1/2/19 9:16	Chair	1200	Mastercard	Sean Davis	Shavano ...	TX	United St...	1/2/19 8:32	1/5/19 9:05	29.42389	-98.49333	3
9	1/5/19 10:08	Chair	1200	Visa	Georgia ...	Eagle	ID	United St...	11/11/18 1...	1/5/19 10:05	43.69556	-116.35306	1
10	1/2/19 14:18	Chair	1200	Visa	Richard P...	Riverside	NJ	United St...	12/9/18 1...	1/5/19 11:01	40.03222	-74.95778	4

Feel free to take several screenshots as needed or make the columns narrower in the sqliteonline output before taking the screenshot (columns can be manually made more narrow by hovering over the lines marked in red below):


```
SELECT * FROM sales;
```

ID	transacti...	product	price	paymen...	name	city	state	country	account...	last_login	latitude	longitude	supplier
1	1/2/19 4:53	Chair	1200	Visa	Bettina S...	Parkville	MO	United St...	1/2/19 4:42	1/2/19 7:49	39.195	-94.68194	2
2	1/2/19 13:08	Chair	1200	Mastercard	Federica ...	Astoria	OR	United St...	1/1/19 16:21	1/3/19 12:32	46.18806	-123.83	3
3	1/4/19 12:56	Couch	3600	Visa	Gerd Win...	Cahaba...	AL	United St...	11/15/18 ...	1/4/19 12:45	33.52056	-86.8025	1
4	1/4/19 13:19	Chair	1200	Visa	Laurence ...	Mickleton	NJ	United St...	9/24/18 1...	1/4/19 13:04	39.79	-75.23806	3
5	1/4/19 20:11	Chair	1200	Mastercard	Fleur Bar...	Peoria	IL	United St...	1/3/19 9:38	1/4/19 19:45	40.69361	-89.58889	2
6	1/2/19 20:09	Chair	1200	Mastercard	Adam Ma...	Martin	TN	United St...	1/2/19 17:43	1/4/19 20:01	36.34333	-88.85028	4
7	1/5/19 2:42	Chair	1200	Diners	Stacy Ro...	New York	NY	United St...	1/5/19 2:23	1/5/19 4:59	40.71417	-74.00639	2
8	1/2/19 9:16	Chair	1200	Mastercard	Sean Davis	Shavan...	TX	United St...	1/2/19 8:32	1/5/19 9:05	29.42389	-98.49333	3
9	1/5/19 10:08	Chair	1200	Visa	Georgia ...	Eagle	ID	United St...	11/11/18 1...	1/5/19 10:05	43.69556	-116.35306	1

For some of these questions, you will need to perform a SUM or AVERAGE in a group-

https://www.khanacademy.org/computing/computer-programming/sql/more-advanced-sql-queries/pt/restricting-grouped-results-with-having

1). 10 points. As you glance through the database, you note that Houston is spelled with a lower case 'h' in at least one case. First of all, write the SQL code to list all values for Houston, capitalized or not capitalized. Begin with **SELECT * FROM sales** to list all fields.

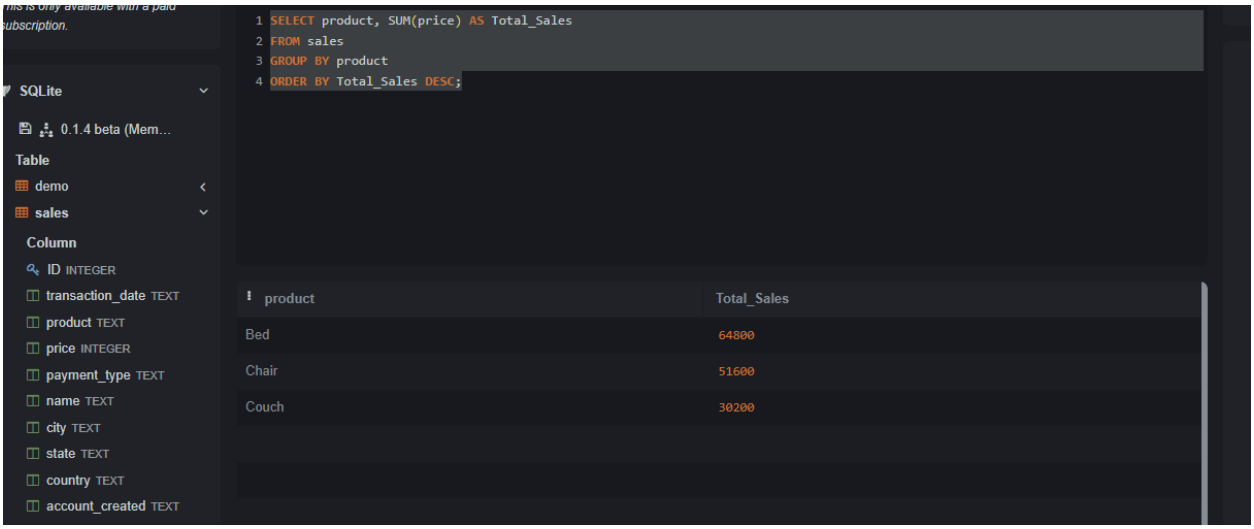
```
SELECT *FROM sales

WHERE city LIKE '%Houston%'
```

Then, write the SQL script to update all lower case instances of houston to Houston. NOTE: use of **WHERE id=** is not permitted. This method is used in the tutorial but is not practical if you had a very large database.

```
UPDATE sales   SET city = REPLACE(city, 'houston', 'Houston')

WHERE city LIKE '%houston%'
```

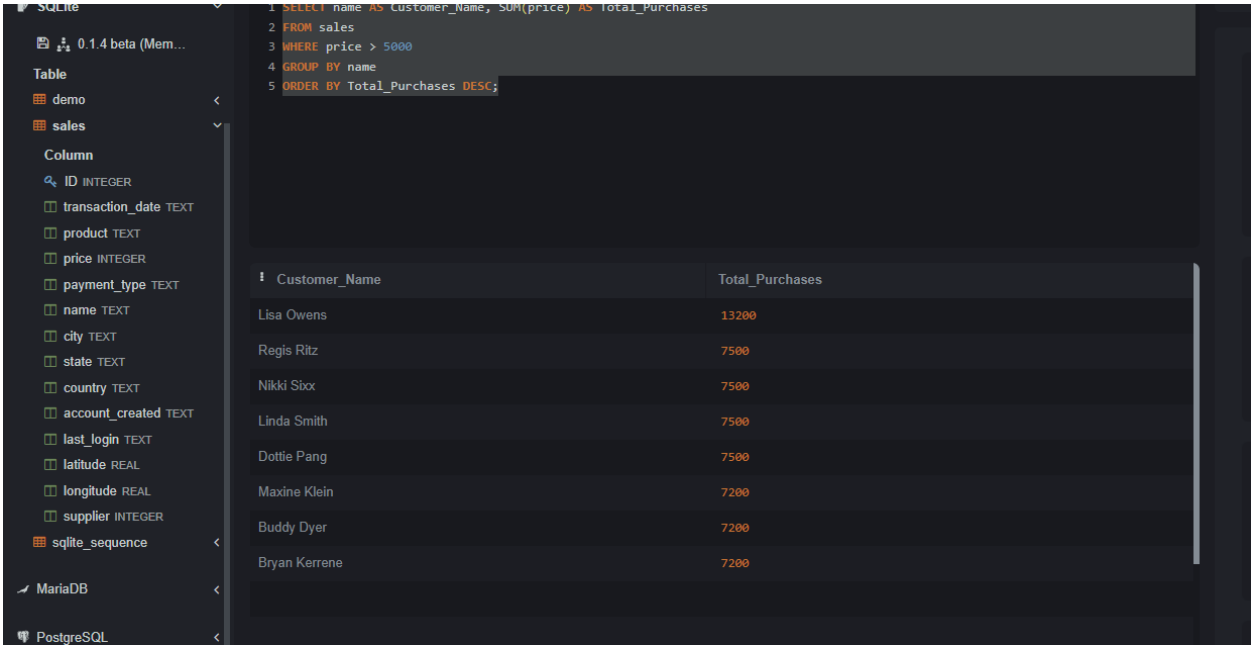
2) 10 points. Write an SQL query to list all products and the total sales in dollars for each product. List products in order from highest total sales to lowest. Use Total_Sales as the label for the field (column) containing the sales data.

```
SELECT product, SUM(price) AS Total_Sales FROM sales

GROUP BY product ORDER BY Total_Sales desc;
```



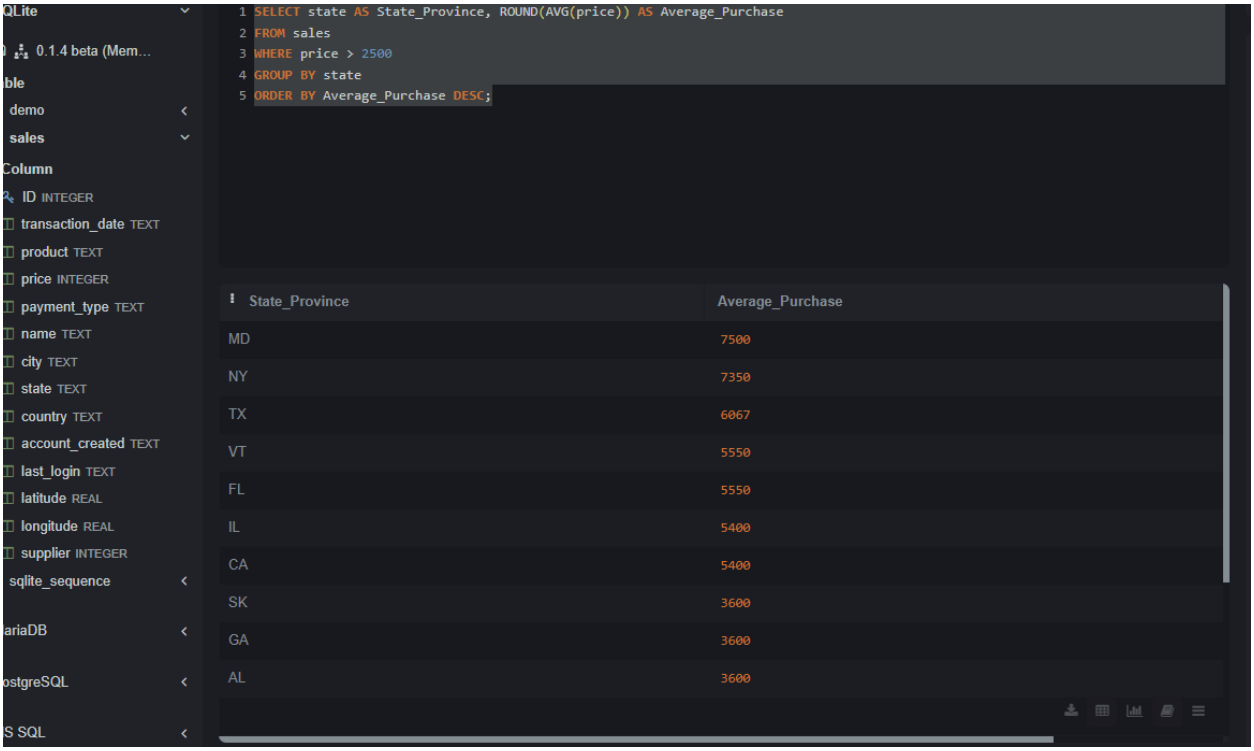
3) 10 points. Write an SQL query to list the customer name and the dollar amount of total purchases. List only names and dollar amounts if total purchases exceed $5000. List in order of total purchases from highest to lowest and use the column titles Customer_Name and Total_Purchases.

```
SELECT name as Customer_Name, SUM(price) as Total_Purchases  FROM sales
WHERE price > 5000  GROUP BY name
ORDER BY Total_Purchases desc;
```



4) 10 points. Write an SQL query to list the states (note Canadian provinces are also included in the database) and the dollar amount of the average purchase. List only states (and provinces) when the average purchase exceeds $2500. List in order of average purchase from highest to lowest and use the column titles State_Province and Average_Purchase. Round the dollar amount of the Average_Purchase so only whole numbers (no decimals) will appear.

```
SELECT state AS State_Province, ROUND(AVG(price)) AS Average_Purchase FROM sales
WHERE price > 2500 GROUP BY state
ORDER BY Average_Purchase desc;
```



5a) NOTE: For all of Q5, this must be an actual situation/scenario that can be executed with an SQL query on the data provided in the assignment. NOT a hypothetical scenario or partial SQL query. If this is unclear, please ask.

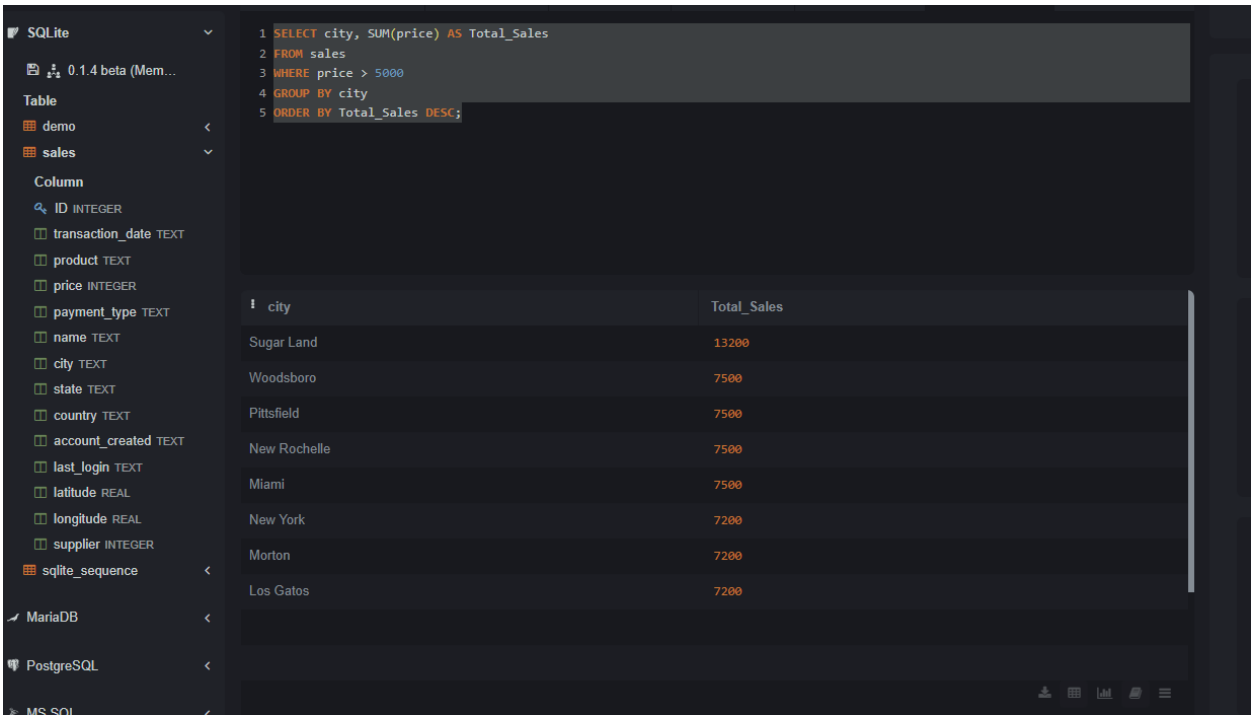
15 points. In 2-3 sentences, given the nature and structure of the **furniturestore.sql** data, describe a situation or scenario (or question that might be answered) that would require use of a CASE statement

The furniture store has sales in many different cities. They are planning on pushing advertising campaigns in there busier cities. We want to determine which cities have total sales over 5000. This will enable us to see which cities are the busiest.

in SQL. (This is a very open question with many correct responses. Think about what the CASE statement in SQL allows us to do.)

5b) Create the SQL query for the scenario presented in part a). This SQL query must be complete, executable and produce output relevant to the scenario presented in part a).

```
SELECT city, SUM(price) AS Total_Sales
FROM sales
WHERE price > 5000
GROUP BY city
ORDER BY Total_Sales desc;
```



6a) 15 points. You have been asked to produce a report that shows all chairs bought on January 12, 2019. In the box below, respond to the following question:

Why does the following SQL query not yield any results (feel free to test it)?

SELECT * FROM sales WHERE product = 'Chair' AND transaction_date = '1/12/19';

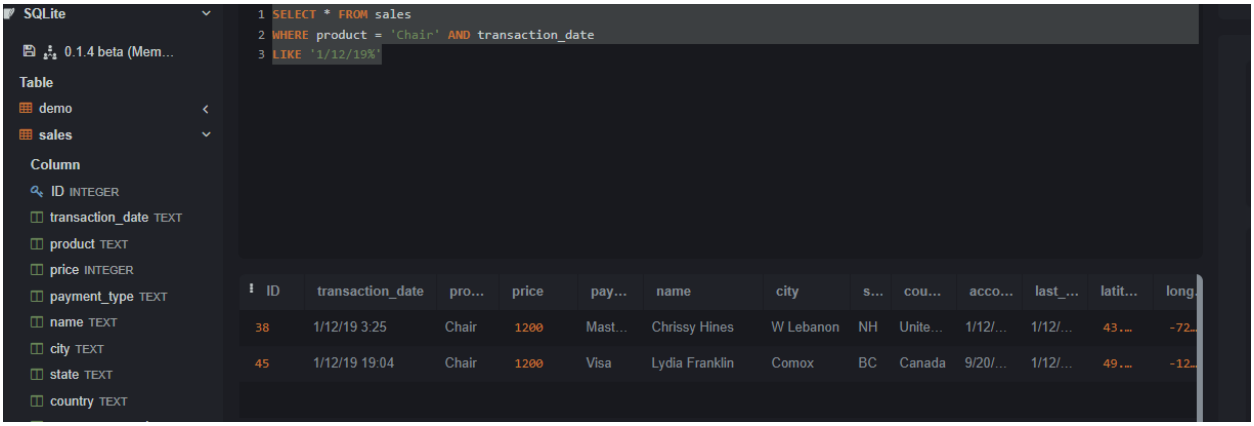
The transaction_date also includes the time in the transaction. The above query is only asked about the date. Since it has to be equal too and the time is not included, they are not equal and there are no results.

6b) Rewrite the query so that it yields the correct results.

SELECT * FROM sales

Where product = 'Chair' AND transaction_date

LIKE '1/12/19%'



PART 2:

Part 2 introduces a second table to the database schema. When creating queries from two tables, in particular when there may be a field that shares a common name, it is safer to construct your queries with the table name AND field name. You may wish to review the "JOINing related tables" tutorial and video on Khan Academy.

7) 5 points. Using an SQL script, add (CREATE) a suppliers table to the database schema, with an autoincrement integer as the primary key (id), and fields for name and country (both text). The primary key (id) will correspond to the supplier field in the sales table (e.g. supplier in sales is a foreign key).

```
CREATE table suppliers (id INTEGER PRIMARY key AUTOINCREMENT, name TEXT, country TEXT,
supplier INTEGER, FOREIGN key (supplier) REFERENCES sales(supplier))
```

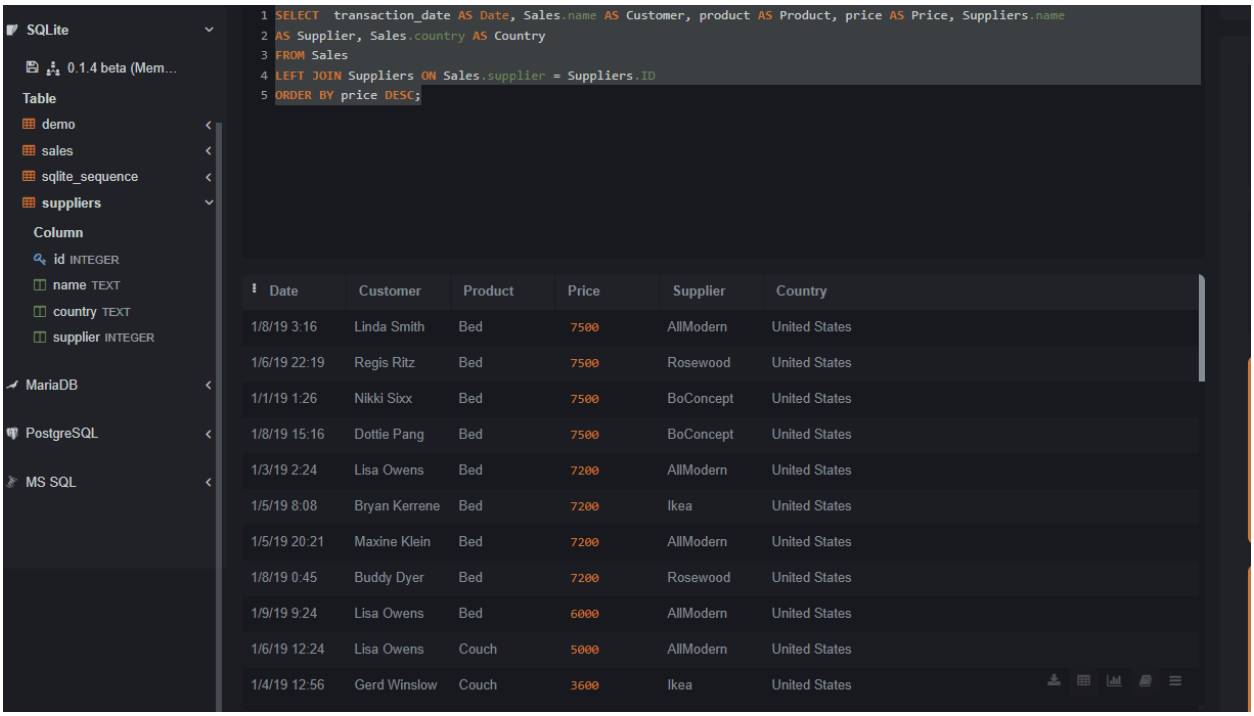
8) 5 points. Using an SQL script, insert the following data into the suppliers table, in the order presented in the table (Reminder: id primary key should have been set to autoincrement when the table was created):

Name	Country
Ikea	Sweden
Rosewood	Thailand
BoConcept	Denmark
AllModern	United States

```
INSERT into suppliers VALUES (1,'Ikea', 'Sweden',1), (2,'Rosewood', 'Thailand',2),
(3,'BoConcept','Denmark',3), (4,'AllModern', 'United States',4)
```

9) 10 points. Write an SQL query to list transaction date, customer name, product, price, supplier name and country (6 columns) ordered by price from highest to lowest. List only items with a price greater than 1500. Construct the SQL query to present field (column) titles as: Date, Customer, Product, Price, Supplier, Country.

```
SELECT  transaction_date AS Date, Sales.name AS Customer, product AS Product, price AS Price, Suppliers.name

as Supplier, Sales.country AS Country

FROM Sales

LEFT JOIN Suppliers ON Sales.supplier = Suppliers.ID

ORDER BY price DESC;
```



10) 10 points. Write an SQL query to list supplier name, country and total sales for that supplier.

```
SELECT suppliers.name as 'Supplier', suppliers.country as 'Country', sum(price) as 'Total Sales'
from sales
JOIN suppliers on sales.supplier=suppliers.supplier
group by suppliers.name
```

Insert (copy and paste) all code from Items 7-10 within this textbox:

```
CREATE table suppliers (id INTEGER PRIMARY key AUTOINCREMENT, name TEXT, country TEXT,
supplier INTEGER, FOREIGN key (supplier) REFERENCES sales(supplier))


INSERT into suppliers VALUES (1,'Ikea', 'Sweden',1), (2,'Rosewood', 'Thailand',2),
(3,'BoConcept','Denmark',3), (4,'AllModern', 'United States',4)


SELECT  transaction_date AS Date, Sales.name AS Customer, product AS Product, price AS Price,
Suppliers.name
as Supplier, Sales.country AS Country
FROM Sales
LEFT JOIN Suppliers ON Sales.supplier = Suppliers.ID
ORDER BY price DESC;

SELECT suppliers.name as 'Supplier', suppliers.country as 'Country', sum(price) as 'Total Sales'
from sales
JOIN suppliers on sales.supplier=suppliers.supplier
group by suppliers.name
```